Zihao Zhao

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Education

University Hospital Aachen — **Aachen, Germany**
Ph.D. student — *Nov.2025–Present*

ShanghaiTech University — **Shanghai, China**
Master Student of Computer Science and Technology — *Sep.2022–Jun.2025*
GPA: 3.54/4.0

Harbin Institute of Technology — **Harbin, China**
Bachelor of Remote Sensing Science and Technology — *Aug.2018–Jul.2022*
GPA: 90.84/100; Rank 1/25

Research Projects

Sep. 2023–Present: **CLIP in medical imaging**

- A comprehensive survey on CLIP in medical imaging was conducted, categorizing existing studies into refined CLIP pre-training and CLIP-driven applications. Accepted by *Medical Image Analysis*.
- We developed knowledge-guided prompt learning based on pre-trained BiomedCLIP, enabling vision models to learn semantic relationships between anatomical variability and biological processes. This property has allowed us to train models for lifespan brain MR image segmentation. Accepted for oral presentation at *MICCAI 2024*.

Feb. 2023–Jan. 2024: **Large language models as diagnostic agents**

- Utilizing LLMs to interpret outputs from multiple CAD models, we improved the decision-making processes and thus proposed ChatCAD. Published in *Communications Engineering*.
- We further advanced this project by proposing ChatCAD+, which was characterized by increased reliability and capability to process inputs from multiple imaging domains. Published in *IEEE TMI*.

Sep. 2022 – Present: **Eye-tracking rectified computer-aided diagnosis**

- We proposed a novel data augmentation approach. Based on gaze heatmaps, it can maintain integrity of crucial visual cues in medical images, enhancing model robustness. The conference paper was accepted by the *NeurIPS 2023 Workshop*, and its extended journal version was accepted by *Pattern Recognition*.
- For the first time, we utilized gaze data to address the issue of false-negatives in medical contrastive learning, with gaze similarity evaluation schemes tailored for various possible scenarios. Accepted for poster presentation at *AAAI 2024*.
- We extended the *AAAI* paper by developing a more advanced method to preserve spatial-temporal information during gaze similarity evaluation. This approach improves performance and validates the function of gaze data in early alignment between human knowledge and neural networks. Accepted by *IEEE TMI*.

Selected Paper

Google Scholar: https://scholar.google.com/citations?user=Novd9cUAAAAJ
* indicates equal contribution.

- **Z. Zhao**, S. Wang, Q. Wang, and D. Shen, "Mining gaze for contrastive learning toward computer-assisted diagnosis", *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, 38 (7): 7543–7551.

https://doi.org/10.1609/aaai.v38i7.28586.

- **Z. Zhao**, S. Wang, J. Gu, Y. Zhu, L. Mei, Z. Zhuang, Z. Cui, Q. Wang, D. Shen. "ChatCAD+: Towards a Universal and Reliable Interactive CAD using LLMs", *IEEE Transactions on Medical Imaging*, 2024, 43 (11): 3755-3766. https://doi.org/10.1109/TMI.2024.3398350.
- **Z. Zhao**, M. Wang, Z. Cui, S. Wang, Q. Zhou, L. Fan, Q. Wang, D. Shen. "Taming Masked Image Modeling for Chest X-ray Diagnosis by Incorporating Clinical Visual Priors", *Information Processing in Medical Imaging*, 2025. In press.
- **Z. Zhao**, Y. Liu, H. Wu, M. Wang, Y. Li, S. Wang, L. Teng, D. Liu, Z. Cui, Q. Wang, D. Shen. "CLIP in Medical Imaging: A Survey", *Medical Image Analysis*, 2025. In press.
- S. Wang*, **Z. Zhao***, X. Ouyang, T. Liu, Q. Wang, D. Shen. "Interactive Computer-Aided Diagnosis on Medical Image using Large Language Models", *Communications Engineering*, 2024, 3 (133). https://doi.org/10.1038/s44172-024-00271-8.
- S. Wang*, **Z. Zhao***, Z. Shen, B. Wang, Q. Wang, D. Shen. "Improving Self-Supervised Medical Image Pre-Training by Early Alignment with Human Eye Gaze Information", *IEEE Transactions on Medical Imaging*, 2025. https://doi.org/10.1109/TMI.2025.3528965.
- S. Wang*, **Z. Zhao***, Z. Zhuang, X. Ouyang, L. Zhang, Z. Li, C. Ma, T. Liu, D. Shen, Q. Wang. "Learning Better Contrastive View from Radiologist's Gaze", *Pattern Recognition*, 2025, 162: 111350. https://doi.org/10.1016/j.patcog.2025.111350

Skills

Language: Mandarin, English (IELTS 7.5)
Programming: Python, PyTorch, C++

Awards

National Scholarship, ShanghaiTech University	2024
Academic Scholarships, ShanghaiTech University	2022-2024
Soochow Industrial Park Scholarship, Harbin Institute of Technology	2021

Activities

Reviewer for:
- IEEE Transactions on Medical Imaging
- IEEE Journal of Biomedical and Health Informatics
- Pattern Recognition
- Neural Networks

Teaching assistant for:
- BME2106 Medical Big-Data and Artificial Intelligence, Spring 2024

Lab assistant for:
- Collection of EEG and eye-tracking data
- Demonstrating the functionalities of eye-tracker during open days

Invited talk for:
- The 13th National Conference on Bioinformatics and Systems Biology, Haikou, China